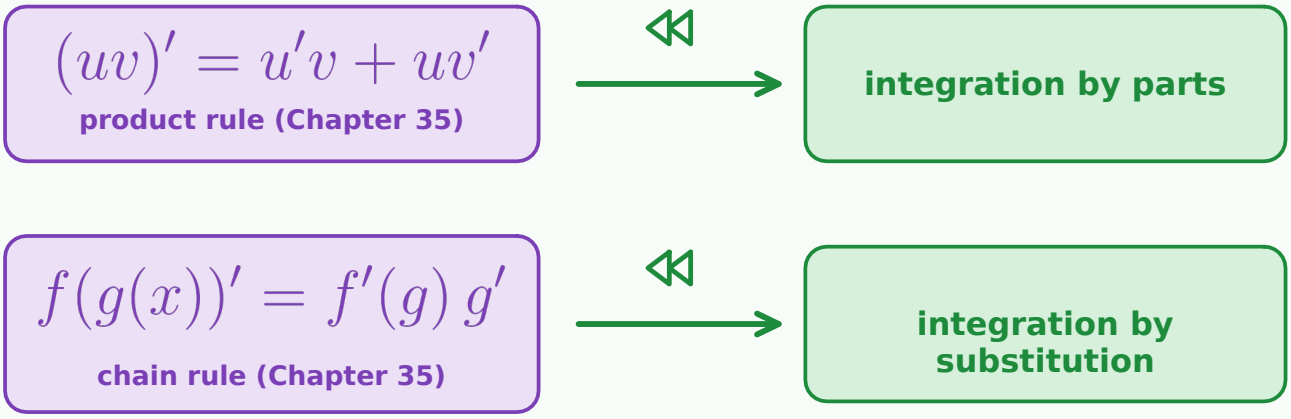
Chapter 37

Integration by parts and substitution

rewind the rules



each technique is a differentiation rule played in reverse

Chapter 36 could integrate single pieces: x^n, e^{kx}, $\sin kx$. It could also add pieces. But what about pieces **multiplied** together, like $x \sin x$, or a function **inside** another, like $(3x+1)^4$?

There's no simple "multiply the integrals" rule. Instead there are two techniques, and each one is a differentiation rule from Chapter 35 run backwards, like pressing rewind.

Goes with **Lecture 17** (recap, products of functions, integration by parts and its derivation, repeated parts, ln x, choosing u, definite integrals by parts, integration by substitution, substitution with limits, the general solution).

⏪ Warm-up: top minus bottom

Lecture 17 opens with a reminder of Chapter 36. Put the top limit in, put the bottom limit in, subtract. A scoreboard helps keep the two apart:

$$\int_0^2 3x^3\,\mathrm{d}x = \left[\tfrac{3}{4}x^4\right]_0^2$$ ← *power up, ÷ new power*

TOP (x = 2)	BOTTOM (x = 0)	
$\frac{3}{4}\times 2^4 = 12$	$\frac{3}{4}\times 0^4 = 0$	$12 - 0 = 12$

Lecture 17's your turn: $\int_1^2 \left(2x^{1/2} - 18x\right)\mathrm{d}x$. The half power is just another power (Chapter 9): up by one to $\frac{3}{2}$, then divide by $\frac{3}{2}$, which is the same as multiplying by $\frac{2}{3}$.

$$= \left[\tfrac{4}{3}x^{3/2} - 9x^2\right]_1^2$$ ← *2 × ⅔ = 4/3, 18 ÷ 2 = 9*

TOP (x = 2)	BOTTOM (x = 1)
$\frac{4}{3}(2)^{3/2} - 36 \approx -32.23$	$\frac{4}{3} - 9 \approx -7.67$

$$-32.23 - (-7.67) = -32.23 + 7.67 = -24.56$$

minus a minus is a plus

⋘ Multiplied pieces: the trap

Adding is easy: $\int (f+g) = \int f + \int g$. So it's tempting to guess that multiplying works the same way. **It doesn't.** Test it on the simplest product there is, $x \times x$:

$$\int x \cdot x \, \mathrm{d}x = \int x^2 \, \mathrm{d}x = \frac{x^3}{3} \quad \longleftarrow \text{the true answer}$$

$$\left(\int x \, \mathrm{d}x\right)\left(\int x \, \mathrm{d}x\right) = \frac{x^2}{2} \cdot \frac{x^2}{2} = \frac{x^4}{4} \quad \times \quad \longleftarrow \text{the guess: wrong power, wrong number}$$

Lecture 17's examples of products: $\int (x^2+7)\sin x \, \mathrm{d}x$, $\int x e^{2x} \, \mathrm{d}x$, $\int f(x)\, g(x) \, \mathrm{d}x$. For these we need **integration by parts**.

Where parts comes from: one rectangle, two pieces

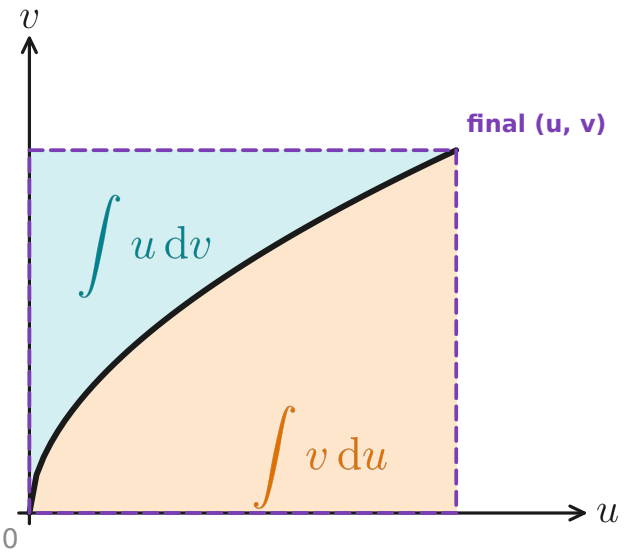
RULE: integration by parts

$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$. Use it for a **product of two different kinds of function** ($x\sin x$, xe^{2x}, $x^2\ln x$).

One factor is u (you will **differentiate** it); the other is $\frac{\mathrm{d}v}{\mathrm{d}x}$ (you will **integrate** it to get v).

Choosing u: LIATE. Whichever factor comes first in Logs, Inverse trig, Algebra (powers of x), Trig, Exponentials is u. A good choice makes the leftover integral $\int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$ simpler than the original.

Where the formula comes from. Chapter 35 pictured $u \times v$ as the area of a rectangle. Now let u and v change together, so that as u grows from 0 to its final size, v grows along a curve. The curve cuts the rectangle into two pieces:



orange: the area under the curve, strips standing on the u axis: ∫ v du

teal: the area beside the curve, strips lying against the v axis: ∫ u dv

together: the whole rectangle, uv

$$\int u\,\mathrm{d}v + \int v\,\mathrm{d}u = uv \quad \longleftarrow \text{the two pieces fill the rectangle}$$

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u \quad \longleftarrow \text{move the orange piece across}$$

So if you want the teal piece but it's hard, you can get it from the rectangle minus the orange piece. That swap is the whole trick. Writing $\mathrm{d}v$ as $\frac{\mathrm{d}v}{\mathrm{d}x}\mathrm{d}x$ gives Lecture 17's version:

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = u\,v - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Lecture 17's algebra route (the same thing without the picture): start from the product rule for $y = uv$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = v\frac{\mathrm{d}u}{\mathrm{d}x} + u\frac{\mathrm{d}v}{\mathrm{d}x}$$ ← Chapter 35

$$u\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} - v\frac{\mathrm{d}u}{\mathrm{d}x}$$ ← subtract to isolate one term

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = \int \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$ ← integrate every term

$$\int \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x = y = uv$$ ← integrating undoes differentiating

The workings box: x sin x

PROCEDURE: integration by parts, step by step

1. Choose u by LIATE; the other factor is $\frac{dv}{dx}$.
2. Differentiate u to get $\frac{du}{dx}$. Integrate $\frac{dv}{dx}$ to get v (no $+C$ yet).
3. Substitute into $uv - \int v \frac{du}{dx}\,dx$.
4. Do the leftover integral. If it is still a product, use parts again.
5. Add a single $+C$ at the end, and check by differentiating.

The formula has four ingredients. Lecture 17 lays them out in a box. You **choose** the top row; the bottom row follows. u gets **differentiated** and $\frac{dv}{dx}$ gets **integrated**:

$u = x$

↓ differentiate

$\frac{du}{dx} = 1$

$\frac{dv}{dx} = \sin x$

↓ integrate

$v = -\cos x$

u = x because it turns into a plain 1 when differentiated.

Now feed the four boxes into the formula, colour for colour:

$$\int u \frac{dv}{dx}\,dx = uv - \int v \frac{du}{dx}\,dx$$

$$\int x \sin x\,dx = x\,(-\cos x) - \int (-\cos x)\,(1)\,dx$$

$$= -x\cos x + \int \cos x\,dx \quad \leftarrow \text{minus times minus}$$

$$= \sin x - x\cos x + C \quad \leftarrow \int \cos x = \sin x \text{ (Chapter 36)}$$

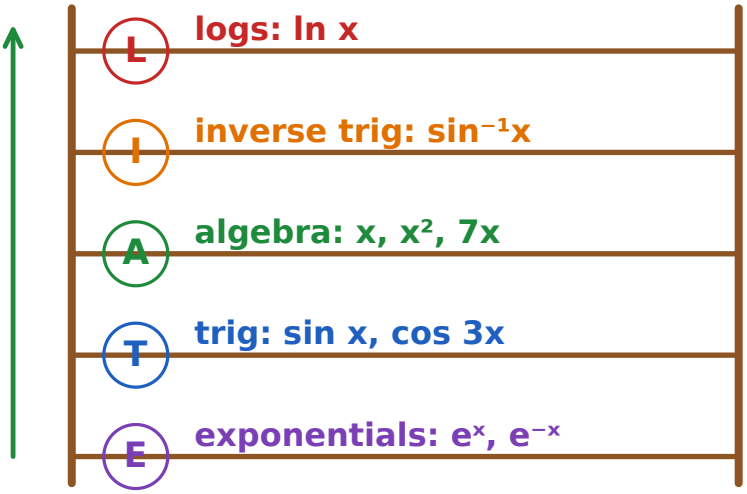
check: $\frac{d}{dx}(\sin x - x\cos x) = \cos x - (\cos x - x\sin x) = x\sin x$ ✓

product rule on x cos x

About the +c. Lecture 17 carries a $+c$ on v and it cancels out ($+cx$ then $-cx$). So leave v without one and add a single $+C$ at the very end. (The slide ends with "$-d$": any constant, plus or minus, means the same thing.)

⏪ Which one is u? The ladder

Lecture 17: you must be able to **differentiate u easily** and **integrate dv/dx easily**, and the leftover integral must be no harder than the one you started with. A handy ordering: climb the ladder, and whichever piece sits **higher** becomes u.



higher rung = make it u

single powers of x are great u's (Lecture 17): differentiate them and they shrink to a number

What happens if you pick the wrong one? Try $x \sin x$ both ways:

u = x (A beats T)

$$\frac{du}{dx} = 1, \quad v = -\cos x$$

$$\text{left over: } \int -\cos x \, dx$$

power of x fell: $x \to 1$

✓ **easier**

u = sin x (wrong way)

$$\frac{du}{dx} = \cos x, \quad v = \frac{x^2}{2}$$

$$\text{left over: } \int \frac{x^2}{2} \cos x \, dx$$

power of x rose: $x \to x^2$

✗ **harder**

⏪ Parts twice: x² cos x

Lecture 17's harder example: $\int x^2 \cos x \, dx$. By the ladder, $u = x^2$ (A beats T).

$u = x^2$ → differentiate → $\frac{du}{dx} = 2x$

$\frac{dv}{dx} = \cos x$ → integrate → $v = \sin x$

$$\int x^2 \cos x \, dx = x^2 \sin x - \int \sin x \cdot 2x \, dx$$

The leftover $\int 2x \sin x \, dx$ is **still a product**. But it's $2\times$ the example we just did, so reuse it:

$$= x^2 \sin x - 2(\sin x - x \cos x) + C$$ ← *Lecture 17's answer*

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$ ← *brackets opened*

A shortcut table. When u is a power of x, parts keeps going until u is differentiated down to 0. Make two columns: **D** (keep differentiating u) and **I** (keep integrating). Join each D to the **next row's** I, with signs $+, -, +, \ldots$ taking turns:

D		I
x^2	+ →	$\cos x$
$2x$	− →	$\sin x$
2	+ →	$-\cos x$
0		$-\sin x$

stop when D hits 0

multiply along each arrow, with its sign:

+ x^2 sin x
− 2x (−cos x)
+ 2 (−sin x)

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$ ← ✓ *the same*

⏪ The invisible 1: integrating ln x

$\ln x$ isn't in Chapter 36's table, and it isn't a product... unless you spot the **invisible 1** that's always there:

$$\int \ln x \, \mathrm{d}x \longrightarrow \int 1 \times \ln x \, \mathrm{d}x$$

ln is top of the ladder, so u = ln x and the 1 is dv/dx

$u = \ln x$	$\dfrac{\mathrm{d}v}{\mathrm{d}x} = 1$
↓ differentiate	↓ integrate
$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{x}$	$v = x$

$$\int \ln x \, \mathrm{d}x = x \ln x - \int x \cdot \frac{1}{x} \, \mathrm{d}x$$

$$= x \ln x - \int 1 \, \mathrm{d}x = x \ln x - x + C$$ ← *x × 1/x = 1*

⏪ Parts with limits

RULE: definite integration by parts (Lecture 17)

$$\int_a^b u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = \Big[uv\Big]_a^b - \int_a^b v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

The uv part gets square brackets and its own top-minus-bottom; the leftover integral keeps the **same limits**. No $+C$ anywhere.

Tidy approach: find the whole indefinite answer first, then put it in one set of square brackets and do top minus bottom once.

Lecture 17: $\int_0^2 xe^{-x}\,\mathrm{d}x$. By the ladder, $u = x$ (A beats E).

$u = x$ → differentiate → $\dfrac{\mathrm{d}u}{\mathrm{d}x} = 1$

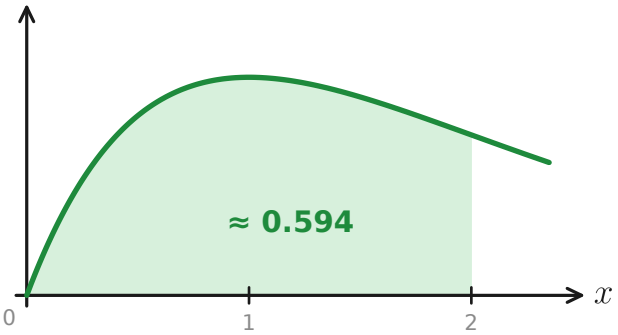
$\dfrac{\mathrm{d}v}{\mathrm{d}x} = e^{-x}$ → integrate → $v = -e^{-x}$

∫e^(−x) = −e^(−x): the k = −1 case of ∫e^(kx) = (1/k)e^(kx)

$$= \Big[-xe^{-x}\Big]_0^2 + \int_0^2 e^{-x}\,\mathrm{d}x = \Big[-xe^{-x} - e^{-x}\Big]_0^2$$

$$= (-2e^{-2} - e^{-2}) - (0 - 1) = 1 - 3e^{-2} \approx 0.594$$

top minus bottom



sanity check: the shape is under 0.37 tall and 2 wide, so an area a bit over half makes sense

Slide slip: Lecture 17's working writes $-xe^{x}$ where it should be $-xe^{-x}$, and e^{2} where it should be e^{-2}. Its final answer $1 - 3e^{-2}$ is right.

⏪ A trig one with limits in radians

Lecture 17: $\int_{\pi/4}^{\pi/2} 7x \sin 3x \, dx$. The limits are in radians (Chapter 27).

$u = 7x$ → differentiate → $\frac{du}{dx} = 7$

$\frac{dv}{dx} = \sin 3x$ → integrate → $v = -\frac{1}{3}\cos 3x$

∫ sin kx = −(1/k)cos kx with k = 3

$$= \left[-\frac{7x}{3}\cos 3x \right] + \int \frac{7}{3}\cos 3x \, dx = \left[-\frac{7x}{3}\cos 3x + \frac{7}{9}\sin 3x \right]_{\pi/4}^{\pi/2}$$

TOP x = π/2

$\cos\frac{3\pi}{2} = 0, \ \sin\frac{3\pi}{2} = -1$

$0 + \frac{7}{9}(-1) = -\frac{7}{9}$

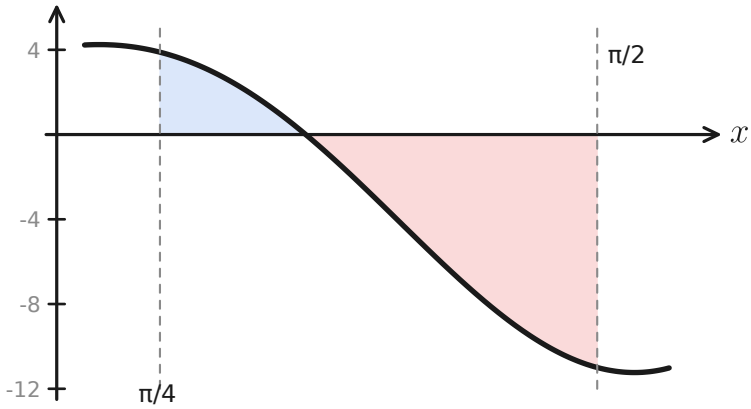
BOTTOM x = π/4

$\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}, \ \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}$

$\frac{7\pi}{12\sqrt{2}} + \frac{7}{9\sqrt{2}} \approx 1.846$

$$-\frac{7}{9} - 1.846 \approx -0.778 - 1.846 \approx -2.62$$ ← top minus bottom

A negative answer? Chapter 36: area below the axis counts negative. The graph shows why:



a small blue piece above the axis (π/4 to π/3), a much bigger red piece below (π/3 to π/2). Net: negative.

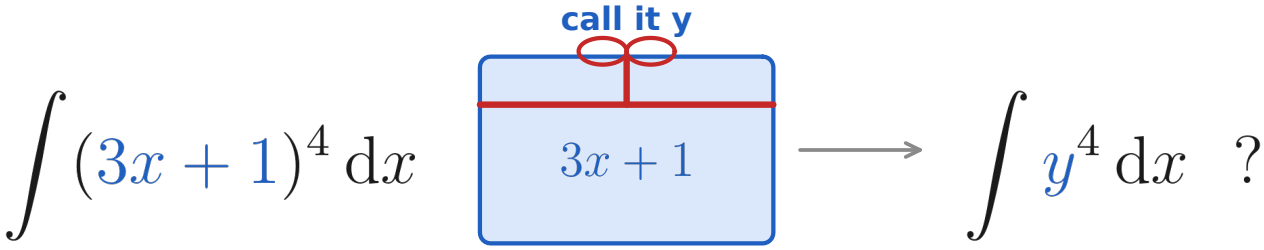
⏪ Substitution: rename the messy chunk

PROCEDURE: integration by substitution

1. **Choose the new letter**: let $y =$ the inside of a bracket, power or function, ideally one whose derivative (give or take a number) also appears in the integral.
2. Differentiate: find $\frac{\mathrm{d}y}{\mathrm{d}x}$, then rearrange to $\mathrm{d}x = \dfrac{\mathrm{d}y}{\mathrm{d}y/\mathrm{d}x}$.
3. Substitute both, so that **every** x disappears (matching x factors cancel). If an x is left over, this choice of y doesn't work.
4. Integrate with respect to y.
5. Indefinite: substitute back, $y =$ (the chunk), and add $+C$. Definite: instead **change the limits** into y-values and never go back to x.

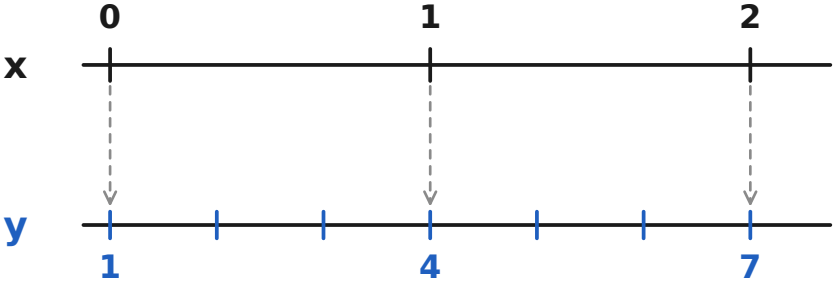
Lecture 17 calls the new letter y, because u is already busy in the parts formula. Many books call it u; the letter doesn't matter as long as it's not x.

Lecture 17: $\int (3x+1)^4 \,\mathrm{d}x$. You could multiply out the fourth power, but that's a pain. Instead, wrap the messy chunk in a parcel and give it a one-letter name:



$$\int (3x+1)^4 \,\mathrm{d}x \longrightarrow \int y^4 \,\mathrm{d}x \quad ?$$

y^4 is easy, but there's a problem: the $\mathrm{d}x$ still says "x". The strips are measured in the wrong units. How do x-steps convert into y-steps? Put two rulers side by side:



one step along x is three steps along y. So dy = 3 dx, which means dx = dy ÷ 3

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \;\Rightarrow\; \mathrm{d}x = \frac{\mathrm{d}y}{3}$$ ← Lecture 17

$$\int (3x+1)^4 \,\mathrm{d}x = \int y^4 \,\frac{\mathrm{d}y}{3} = \frac{1}{3} \cdot \frac{y^5}{5} + C$$ ← now everything is y

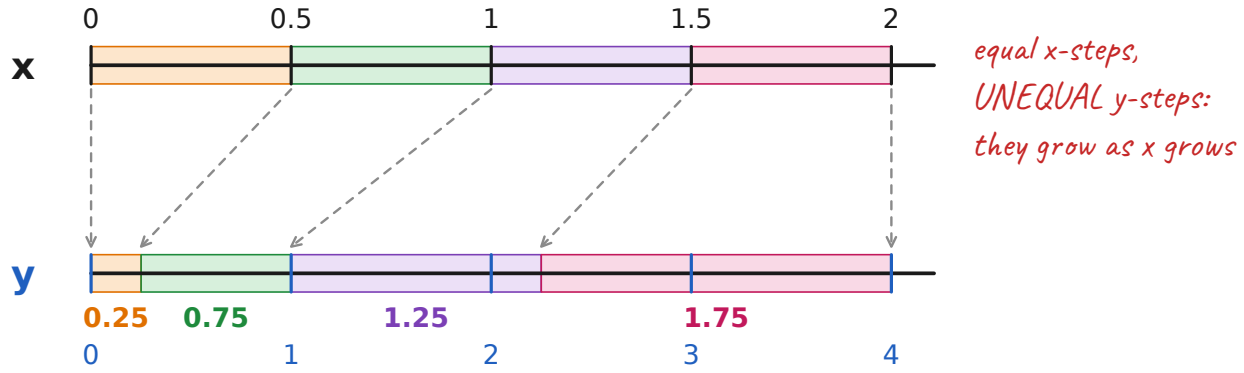
$$= \frac{(3x+1)^5}{15} + C$$ ← unwrap the parcel: y = 3x + 1

check: $\dfrac{\mathrm{d}}{\mathrm{d}x}\dfrac{(3x+1)^5}{15} = \dfrac{5(3x+1)^4 \times 3}{15} = (3x+1)^4$ ✓

chain rule

⏪ The partner outside the bracket

First, why a partner is needed at all. With $y = 3x + 1$ the rulers matched evenly: every x-step was 3 y-steps. Try $y = x^2$. Mark equal x-steps of 0.5 and see where they land on a y-ruler:



Each y-step is $2x \times 0.5$, where x is the middle of that strip: $2(0.25)(0.5) = 0.25$, $2(0.75)(0.5) = 0.75$, and so on. So $\mathrm{d}y = 2x\,\mathrm{d}x$: the exchange rate between the rulers **changes as you go**, and it's $\frac{\mathrm{d}y}{\mathrm{d}x}$. When you swap $\mathrm{d}x$ for $\mathrm{d}y$ you must bring that rate along, and it only disappears if a matching x is sitting outside to cancel it. That's the partner.

Lecture 17: $\int x \sin(2x^2)\,\mathrm{d}x$. The messy chunk is $2x^2$, so let $y = 2x^2$. Its derivative is $4x$... and look: an x is already sitting outside, waiting. Colour them:

$$\int x\,\sin(2x^2)\,\mathrm{d}x, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = 4x \;\Rightarrow\; \mathrm{d}x = \frac{\mathrm{d}y}{4x}$$

$$= \int \cancel{x}\,\sin y\,\frac{\mathrm{d}y}{4\cancel{x}} = \frac{1}{4}\int \sin y\,\mathrm{d}y \quad \leftarrow \text{the partners cancel: only } y \text{ is left}$$

$$= -\frac{1}{4}\cos y + C = -\frac{\cos(2x^2)}{4} + C \quad \leftarrow \int \sin = -\cos$$

Slide slip: Lecture 17 writes $+\frac{\cos(2x^2)}{4}$. Differentiate it and you get $-x\sin(2x^2)$, the wrong sign, because $\int \sin y\,\mathrm{d}y = -\cos y$.

Lecture 17's your turn: $\int t\sqrt{t^2+1}\,\mathrm{d}t$. Chunk t^2+1, derivative $2t$, partner t outside:

$$y = t^2+1, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 2t, \quad \mathrm{d}t = \frac{\mathrm{d}y}{2t}$$

$$\int t\sqrt{y}\,\frac{\mathrm{d}y}{2t} = \frac{1}{2}\int y^{1/2}\,\mathrm{d}y \leftarrow \sqrt{}\text{ is the power } \tfrac{1}{2}$$

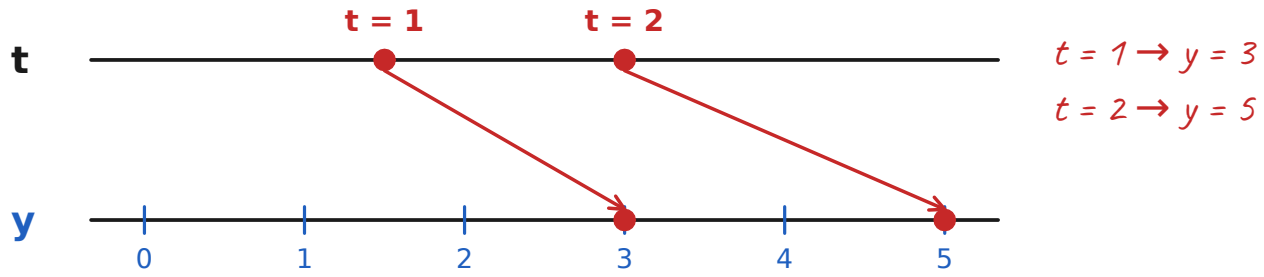
$$= \frac{1}{2}\cdot\frac{2}{3}y^{3/2} + C = \frac{1}{3}(t^2+1)^{3/2} + C$$

The spotting test. Substitution works when the derivative of the chunk (give or take a number) is sitting outside as a partner. No partner, no cancelling: $\int \sin(2x^2)\,\mathrm{d}x$ can't be done this way, because $\frac{\mathrm{d}x}{4x}$ would leave an x stranded among the y's.

⏪ Substitution with limits: change the limits too

Lecture 17: $\int_1^2 \frac{3}{(2t+1)^2}\,\mathrm{d}t$. Let $y = 2t+1$, so $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and $\mathrm{d}t = \frac{\mathrm{d}y}{2}$.

The limits 1 and 2 are values of t. Once everything is in y, the limits must be y values too. Convert them with the same rule, like converting pounds into pence:



$$\int_1^2 \frac{3}{(2t+1)^2}\,\mathrm{d}t = \int_3^5 \frac{3}{y^2}\cdot\frac{\mathrm{d}y}{2} = \frac{3}{2}\int_3^5 y^{-2}\,\mathrm{d}y$$

new limits, new letter

$$= \frac{3}{2}\Big[-\frac{1}{y}\Big]_3^5 = \frac{3}{2}\Big(-\frac{1}{5}+\frac{1}{3}\Big) = \frac{3}{2}\cdot\frac{2}{15} = 0.2$$

no need to swap back to t

The general pattern (Lecture 17). Every substitution example had a chunk $g(x)$ **and its derivative** $g'(x)$ multiplied together:

$$\int_a^b f\big(g(x)\big)\,g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(y)\,\mathrm{d}y \quad \longleftarrow\ y = g(x)$$

That's the **chain rule backwards**: Chapter 35's chain rule makes exactly this shape (outer derivative × inner derivative). In $x\sin(2x^2)$, x is a quarter of the derivative of $2x^2$; in $t\sqrt{t^2+1}$, t is half the derivative of t^2.

⏩ Shortcut: a linear inside, ax + b

RULE: integrating f(ax + b)

If $\int f(x)\,dx = F(x) + C$, then $\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$.

In words: integrate as if the bracket were plain x, then **divide by** a, the number in front of x.

Only when the inside is linear ($ax + b$). For an inside like $x^2 + 1$ you can't divide by its derivative $2x$: use substitution with a partner, or expand.

It is substitution done in your head: $y = ax + b$ gives $dx = \frac{dy}{a}$, and that $\frac{1}{a}$ comes straight out front. Lecture 17's first example agrees: $\int (3x+1)^4\,dx = \frac{1}{3}\cdot\frac{(3x+1)^5}{5} = \frac{(3x+1)^5}{15}$.

$$\int (3x+1)^5\,dx = \frac{1}{3}\cdot\frac{(3x+1)^6}{6} = \frac{(3x+1)^6}{18} + C$$

$a = 3$

$$\int \cos(2x)\,dx = \tfrac{1}{2}\sin(2x) + C \quad \leftarrow a = 2$$

$$\int e^{4x}\,dx = \tfrac{1}{4}e^{4x} + C \quad \leftarrow a = 4$$

$$\int \frac{1}{2x+3}\,dx = \tfrac{1}{2}\ln|2x+3| + C \quad \leftarrow 1/x \to \ln|x|,\ a = 2$$

$$\int \sqrt{4x-1}\,dx = \frac{1}{4}\cdot\frac{(4x-1)^{3/2}}{3/2} = \frac{(4x-1)^{3/2}}{6} + C$$

$a = 4$

✗ Common mistakes

✗ $\int (3x+1)^5 \mathrm{d}x = \frac{(3x+1)^6}{6}$ ✓ $\frac{(3x+1)^6}{18} + C$

divide by the 3 in front of x as well

✗ $\int e^{4x}\,\mathrm{d}x = 4e^{4x}$ ✓ $\int e^{4x}\,\mathrm{d}x = \frac{1}{4}e^{4x} + C$

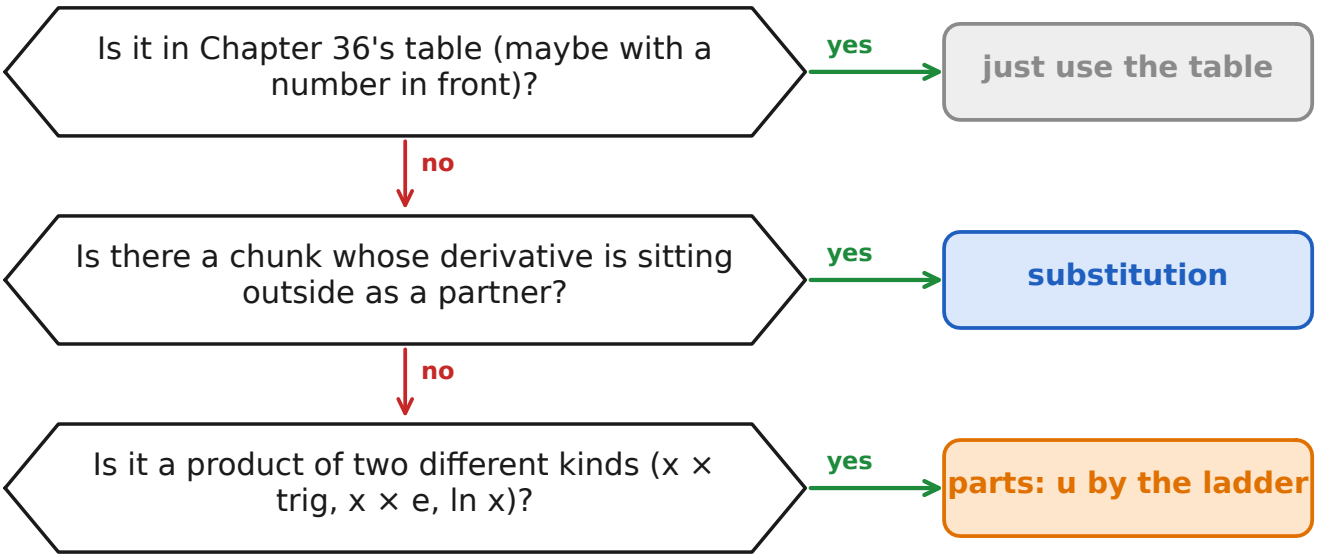
integrating divides by a; differentiating multiplies

✗ $\int (x^2+1)^5\,\mathrm{d}x = \frac{(x^2+1)^6}{6 \cdot 2x}$ ✓ expand first

the shortcut needs a linear inside

⏪ Parts, substitution, or neither? A sorting machine

Ask the questions in this order. Stop at the first yes.



no to all three: try rewriting it first (expand, split, use an identity)

Sort these (answers checked by differentiating):

$\int x\,e^{x^2}\,\mathrm{d}x$	**substitution: y = x², partner x**	$\frac{1}{2}e^{x^2}+C$
$\int x\,e^{x}\,\mathrm{d}x$	**parts: u = x**	xe^x-e^x+C
$\int e^{3x}\,\mathrm{d}x$	**table**	$\frac{1}{3}e^{3x}+C$
$\int \cos x\,\sin^4 x\,\mathrm{d}x$	**substitution: y = sin x, partner cos x**	$\frac{1}{5}\sin^5 x+C$
$\int x\ln x\,\mathrm{d}x$	**parts: u = ln x (L beats A)**	$\frac{x^2}{2}\ln x-\frac{x^2}{4}+C$

And the big picture: **integration by parts undoes the product rule; substitution undoes the chain rule.** Both are the undo button (Chapter 36) for a Chapter 35 rule.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Integration by parts	$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$ for a product of two different kinds of function
Choosing u	LIATE: Logs, Inverse trig, Algebra, Trig, Exponentials; the earlier one is u
Parts with limits	$\int_a^b u\frac{dv}{dx}\,dx = \Big[uv\Big]_a^b - \int_a^b v\frac{du}{dx}\,dx$ same limits on both parts
Parts twice	if the leftover is still a product, use parts again (or the D/I table)
ln x	$\int \ln x\,dx = x\ln x - x + C$ parts with $u = \ln x$, $\frac{dv}{dx} = 1$
Substitution	let y = inside chunk; $dx = \frac{dy}{dy/dx}$; every x must disappear; integrate; substitute back
Substitution with limits	$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(y)\,dy$ change the limits; don't go back to x
Linear inside	$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$ only for $ax+b$
Which method?	in the table: use it. Chunk with its derivative outside: substitution. Product of different kinds: parts. Otherwise rewrite first

⏪ Your turn

Check every answer by differentiating it. Answers in grey.

Integration by parts

1. ∫ x cos x dx
2. ∫ x e^(2x) dx
3. ∫ x² e^x dx (use the D/I table)
4. ∫ from 1 to e of ln x dx

1. x sin x + cos x + C 2. ½x e^(2x) − ¼e^(2x) + C 3. e^x(x² − 2x + 2) + C 4. 1

Integration by substitution

5. ∫ (2x − 5)⁶ dx
6. ∫ 6x (x² + 4)⁵ dx
7. ∫ cos x · e^(sin x) dx
8. ∫ from 0 to 1 of 2x (x² + 1)³ dx (change the limits)

5. (2x − 5)⁷/14 + C 6. (x² + 4)⁶/2 + C 7. e^(sin x) + C 8. 15/4 (limits become 1 and 2)

Which technique?

9. ∫ x sin(x²) dx
10. ∫ x sin x dx
11. ∫ sin 5x dx

9. substitution: −½cos(x²) + C 10. parts: sin x − x cos x + C 11. table: −⅕cos 5x + C

Linear inside shortcut

12. ∫ (5x − 2)³ dx
13. ∫ sin(3x + 1) dx
14. ∫ e^(−2x) dx
15. ∫ 1/(4x + 1) dx

12. (5x − 2)⁴/20 + C 13. −⅓cos(3x + 1) + C 14. −½e^(−2x) + C 15. ¼ ln|4x + 1| + C

Find the unknown

16. ∫ from 0 to k of x e^x dx = 1, with k > 0. Find k.
17. ∫ (ax + 1)⁴ dx = (ax + 1)⁵/10 + C. Find a.
18. ∫ from 0 to k of 2x(x² + 1)³ dx = 255/4, with k > 0. Find k.
19. dy/dx = x e^x and y = 3 when x = 0. Find y.
20. Find the smallest k > 0 with ∫ from 0 to k of cos 2x dx = ½.

16. k = 1 ([x e^x − e^x] gives (k − 1)e^k + 1 = 1) 17. a = 2 (1/(5a) = 1/10) 18. k = √3 (((k² + 1)⁴ − 1)/4 = 255/4, so k² + 1 = 4) 19. y = (x − 1)e^x + 4 (−1 + C = 3) 20. k = π/4 (½ sin 2k = ½, so 2k = π/2)